MedTourEasy

Corporate Product Requirements Document

Project: Patient Conversion & Provider Performance Intelligence

Company	MedTourEasy
Department	Healthcare Analytics / Data Science
Project Duration	6 Weeks
Primary Output	Management-ready analytics workbook, SQL dataset, ML risk scoring pipeline, and executive recommendation memo
Project Owner	Analytics Manager
Candidate Role	Data Science Intern

1. Executive Summary

MedTourEasy requires a patient journey intelligence analysis that identifies where international healthcare leads lose momentum, which providers convert most efficiently, and which patients require proactive intervention. The intern will build an end-to-end Python analytics solution using structured patient, provider, communication, treatment, and reference datasets.

The final output must support management decisions on provider prioritisation, coordinator workflow improvement, conversion recovery, and measurable service-quality interventions.

2. Business Problem

- Patient data is fragmented across inquiry, communication, consultation, quote, treatment, and follow-up stages.
- Provider teams need earlier signals of disengagement risk before patients stop responding.
- Management needs reliable KPI definitions, reconciled funnel numbers, and actionable operational recommendations.
- MedTourEasy needs repeatable analytics assets that can be reused across future partner-provider review cycles.

3. Objectives and Non-Goals

Area	In Scope	Out of Scope
Analytics	Funnel analysis, EDA, segmentation, hypothesis testing, provider benchmarking, and forecasting-ready KPI logic.	Clinical diagnosis, treatment recommendation, or claims of medical efficacy.
Machine Learning	Predict disengagement risk and completed-treatment probability using interpretable supervised models.	Production deployment, automated patient outreach, or regulated decision support.
Data Engineering	Clean CSVs, create SQL tables, define data dictionary, and document reproducible processing steps.	Connection to live CRM, hospital HIS, or PHI systems.
Communication	Executive dashboard narrative, concise recommendations, assumptions log, and QA checklist.	Marketing collateral or public-facing claims.

4. Dataset Pack

File	Rows	Key Fields	Primary Use
patient_journey.csv	1,500	patient_id, inquiry_date, country, treatment_category, provider_id, response_time_hours, engagement_score, consultation_booked, quote_shared, treatment_completed, drop_off_stage, revenue	Funnel analysis, cohort analysis, risk scoring, segmentation

provider_master.csv	30	provider_id, region, specialty_area, response SLA, conversion rates, satisfaction, pricing tier	Provider benchmarking and partner performance scoring
communication_logs.csv	6,025	communication_id, patient_id, timestamp, channel, interaction_type, response_sent, response_delay_hours, follow_up_status	Communication bottleneck analysis and follow-up effectiveness
treatment_outcomes.csv	554	treatment_date, length_of_stay, revenue, cost, margin, readmission flag, follow-up rating	Outcome and unit economics analysis
country_reference.csv	12	country, region, travel lead time, visa requirement, baseline SLA, priority score	Market segmentation and SLA benchmarking
mte_patient_intelligence.db	5 tables	Same data in relational format	SQL joins, aggregation, and reproducible query work

5. Required Analytical Workstreams

- **Data Quality & Governance**

Profile missing values, duplicates, date validity, categorical consistency, revenue/cost reconciliation, and assumption log.

- **Exploratory Analysis**

Create funnel metrics, drop-off rates by stage, response-time distributions, conversion by provider, treatment category, country, source channel, and urgency.

- **Statistical Testing**

Test whether response SLA breaches are associated with consultation booking, whether engagement scores differ by lead source, and whether provider conversion differences are statistically meaningful.

- **Feature Engineering**

Create SLA breach, days-to-treatment, communication coverage, high-value case, market-priority, provider-performance, and interaction-density features.

- **Predictive Modeling**

Build baseline logistic regression and tree-based models for drop-off risk and treatment completion; compare ROC-AUC, precision, recall, calibration, and explainability.

- **Unsupervised Learning**

Segment patients/providers using clustering; describe segments in business language and recommend actions.

- **SQL Analytics**

Create at least eight SQL queries covering joins, aggregations, windows/ranks, cohort conversion, and provider scorecards.

- **Executive Communication**

Prepare a dashboard-ready KPI framework and final memo with quantified recommendations, risks, and next steps.

6. Functional Requirements

ID	Requirement	Acceptance Criteria	Primary Skill Tested	Priority

FR-01	Load and validate all CSV and SQL tables	Notebook runs from clean environment; row counts and schema checks are printed	Importing data, SQL, pandas	P0
FR-02	Produce a reconciled funnel table	All funnel stages reconcile to 1,500 patient records; no double-counted drop-offs	pandas, groupby, data QA	P0
FR-03	Build provider performance scorecard	Provider ranks include conversion, response SLA, follow-up, satisfaction, and margin indicators	joins, aggregation, visualization	P0
FR-04	Create predictive risk model	At least two models compared with train/test split, preprocessing, metrics, and feature importance	scikit-learn, preprocessing	P0
FR-05	Create patient/provider segments	Cluster outputs include profile table and recommended operating action per segment	scaling, clustering, EDA	P1
FR-06	Develop SQL analysis layer	SQL script includes minimum eight business queries with comments and outputs	relational database SQL	P1
FR-07	Package reusable code	Functions stored in a small Python package or scripts folder with README and requirements.txt	Git/package structure	P1
FR-08	Management recommendation memo	Recommendations are quantified, prioritised, and tied to evidence from the analysis	business communication	P0

7. Deliverables

- 01_data_quality_report.ipynb
- 02_exploratory_patient_funnel_analysis.ipynb
- 03_sql_provider_scorecard.sql plus saved outputs
- 04_modeling_dropoff_and_completion.ipynb
- 05_segmentation_and_recommendations.ipynb
- Executive dashboard workbook or notebook visuals
- Final PRD-style analytics memo: problem, method, findings, recommendations, risks, and next steps
- Git repository structure with README, requirements.txt, and reproducibility notes

8. Evaluation Rubric

Category	Weight	Expected Standard	Common Failure to Avoid
Data accuracy and reconciliation	20%	Counts, joins, financial totals, and stage definitions reconcile.	Double-counting unresolved/drop-off cases.
Analytical depth	20%	Findings go beyond charts and explain drivers, segments, and operational levers.	Generic observations without quantified impact.

Modeling quality	20%	Proper preprocessing, validation, comparison, error analysis, and explainability.	Reporting only accuracy or training-set results.
Business usefulness	20%	Recommendations map to owners, expected impact, and measurable KPIs.	Unprioritised or unsupported recommendations.
Communication and reproducibility	20%	Clean notebooks, SQL, visual design, assumptions log, and executive narrative.	Messy notebooks, missing README, or inconsistent claims.

9. Suggested Six-Week Timeline

Week	Focus	Milestone
1	Data onboarding, schema review, quality checks	Validated data dictionary and QA report
2	EDA, funnel, provider and communication analysis	Insight notebook with reconciled KPI tables
3	SQL scorecards and hypothesis testing	SQL script and statistical testing summary
4	Feature engineering and supervised ML	Baseline and tuned model comparison
5	Segmentation, dashboard visuals, recommendations	Segment profiles and dashboard-ready visuals
6	Final packaging and management narrative	Final memo, code package, and presentation-ready recommendations

10. KPI Definitions

- Consultation Conversion Rate = consultation_booked / inquiries
- Treatment Completion Rate = treatment_completed / inquiries
- Quote-to-Treatment Rate = treatment_completed / quote_shared
- SLA Breach Rate = share of inquiries with response_time_hours above country baseline
- Communication Coverage = patients with at least one completed follow-up interaction / total patients
- Gross Margin = actual_revenue_inr - service_cost_inr
- Provider Intelligence Score = weighted composite of conversion, SLA adherence, follow-up completion, satisfaction, and margin

11. Data Ethics and Privacy Requirements

- Do not infer or recommend clinical treatment decisions.
- Do not use personal identifiers outside the provided patient_id keys.
- Report model limitations and avoid causal claims unless supported by test design.
- Maintain an assumptions log for ambiguous business rules and explain any excluded rows.

12. Final Acceptance Checklist

- All tables load successfully from both CSV and SQLite.
- All funnel metrics reconcile to base population.
- Provider scorecard uses consistent weights and documented formulas.
- Model outputs include metrics, confusion matrix, threshold discussion, and feature importance.
- Final recommendations include owner, action, expected impact, KPI, and implementation risk.
- Submission package has no unrelated files and is ready for LMS upload.